

Sahil Kumar

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Professional Summary

Pre-final year dual-degree student (B.Tech in Electronics and Communication Engineering at NIT Uttarakhand and BS in Data Science at IIT Madras) with strong experience in designing and deploying AI/ML systems. Skilled in natural language processing, recommendation models, and multimodal AI applications. Hands-on expertise with TensorFlow, and cloud-based microservice deployment using Flask/FastAPI and Docker. Proven ability to build data-driven solutions, from preprocessing and model training to scalable API integration and visualization.

Education

National Institute of Technology, Uttarakhand (NITUK) B.Tech in Electronics and Communication Engineering	Aug 2023 – Present
Indian Institute of Technology, Madras (IITM) BS in Programming and Data Science and its Applications	Sep 2023 – Present
Shri Balwant Singh Sr. Sec. School, Hathras (CBSE) Class 12: 95% — Class 10: 90%	2020 – 2022

Projects

Data Analyst Agent – Deployed an LLM-powered API that accepts tasks and data files via POST requests, performs sourcing, preparation, analysis, and visualization, and returns structured JSON responses. Built with Python, FastAPI, Pandas, Matplotlib, and DuckDB.

Vehicle Parking Management App

- Developed a full-stack web application for multi-user parking management with separate Admin and User dashboards.
- Implemented secure authentication and role-based access control for administrators and users.
- Built CRUD operations for parking lots and spots; added auto-generation of spots on lot creation.
- Integrated live analytics dashboard using Chart.js for earnings, daily revenue, and usage insights.
- Enabled real-time search and filtering for reservations by user, vehicle number, lot, or spot.
- Created reservation system with cost auto-calculation and user spending analytics.
- Designed a responsive front end using HTML, CSS, JavaScript, and dynamic templates via Jinja2.

Virtual Teaching Assistant (IITM) – Developed a RAG-based chatbot using LangChain and ChromaDB for automated Q&A.

YouTube Video Summarizer – Using Langchain and GPT for transcription and summarization, reducing watch time.

Positions of Responsibility

Student Placement Representative (SPR), ECE Dept., NIT Uttarakhand (2023–27).

Core Member, Bureau of Indian Standards (BIS) Student Chapter, NIT Uttarakhand.

Certifications

IIT Madras Foundation Level – Programming and Data Science.

Extra-Curricular Activities

Core Team Member – NIDAR 2025 Drone Competition.

Active Coordinator – IIT Madras House Initiatives.

Soft Skills

Problem-solving, Logical reasoning, Communication, Collaboration, Adaptability, Eagerness to learn.